

The Foundation: $(\mathbb{S}, \tau)$

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$(\mathbb{S}, \tau)$ is the foundation. A state, a temperature, a specific physical realization — these are scaffolding, kept only until something more fundamental supplies or replaces them. This document states the distinction plainly rather than hedging it.

1. The algebra, unconditionally

[DEF] $\mathbb{S} = \mathbb{R}_\varepsilon[(\mathbb{Z}_2)^4]$: the sixteen-dimensional real Cayley–Dickson algebra, the specific cocycle produced by iterated doubling from $\mathbb{R}$. Unital, flexible, quadratic, power-associative, positive-definite norm. Not associative, not alternative, has zero divisors — starting data, not omissions. $\text{Aut}(\mathbb{S}) = G_2 \times S_3$.

[DEF] τ : an involution in the S_3 factor, splitting $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$: $\mathbb{O} = \text{Fix}(\tau)$, $\mathbb{O}e_8 = \ker(\tau + 1)$. Three conjugate splits exist, related by the order-three μ ; the choice among them is symmetric — an orbit coordinate, not a further commitment.

[FACT] The mediator direction used below is, by the same logic, an orbit coordinate: G_2 acts transitively on the unit sphere in $\text{Im}(\mathbb{O})$, so any choice of mediator is related to any other by an automorphism, and none is a further commitment.

[DEF] Fixing a mediator $e \in \text{Im}(\mathbb{O})$, $|e| = 1$: a symplectic form $\sigma(x, y)$ as the e -component of $[x, y]$; a complex structure $J := L_e$; a metric $g(x, y) := \sigma(x, Je)$, compatible with both. This is the standard compatible-triple construction of symplectic geometry, and it is not imported: it recovers $\text{Im}(\mathbb{O})$'s own composition structure exactly, since for imaginary x, y , $xy = -\langle x, y \rangle + x \times y$, so g built this way reproduces $-\text{Re}(xy)$ up to the normalization σ itself carries.

[FACT, proven] The full geometric skeleton (σ, J, g) is fixed by $(\mathbb{S}, \tau, \text{mediator})$ alone. Nothing beyond this triple of choices — the algebra, the split, one orbit-coordinate direction — is needed to define it, and nothing about it is negotiable once those choices are made.

2. What $\mathbb{S}$ already supplies for both sectors, from one multiplication table

[FACT, verified] For $x, y \in \text{Im}(\mathbb{O})$, the odd part of the τ -split satisfies, under $\mathbb{S}$'s own multiplication, the exact Clifford relation:

$$\{(0, x), (0, y)\} = -2\langle x, y \rangle \cdot 1.$$

Verified directly: orthogonal x, y anticommute exactly (zero to machine precision); $x = y$ gives exactly $-2\|x\|^2$. This is not a Clifford structure built abstractly on top of $\mathbb{S}$'s quadratic form and grafted on afterward — it is what $\mathbb{S}$'s own multiplication already does to the odd half of its own τ -grading. The even part's commutator, $[x, y]$ for $x, y \in \text{Im}(\mathbb{O})$, is what §1's symplectic form σ is built from. One multiplication table, one involution, two grading-halves: the commutator structure feeding the geometric skeleton, and the anticommutator structure matching the Clifford relations a fermionic construction needs, both native to $\mathbb{S}$, not two separately-sourced ingredients.

[FACT, proven] No nonzero real element of $\mathbb{S}$ can supply a self-annihilating creation operator by serving *directly* as the product — shown two independent ways: strictly positive length makes squaring-to-zero impossible for the element itself; a trace identity on left-multiplication operators closes the same route at the operator level, confirmed directly ($L_x^2 = -\|x\|^2 \cdot \text{Id}$, eigenvalues purely imaginary, the opposite of nilpotent). This does not block a fermionic construction — the standard route needs only the Clifford relation above, which §2’s own fact already supplies unconditionally — but it closes the naive route of building a creation operator from a single element’s own self-product.

[FACT, proven] No finite construction can produce $[X, P] = i \cdot \text{Id}$ from finitely many fermionic modes: $\text{tr}(AB - BA) = 0$ identically for any operators on any finite-dimensional space, while $\text{tr}(i \cdot \text{Id}) = in \neq 0$ for finite n . Any finite number of fermionic modes gives a finite-dimensional Fock space. This closes a specific, tempting unification route — building the canonical pair directly as a fermion bilinear — without closing the weaker, structural unification §2’s own fact already establishes: shared origin, not shared construction.

3. What is scaffolding, named as such

[FACT, proven] $(\mathbb{S}, \tau, \text{mediator})$ forces (σ, J, g) and nothing more. Talking about a specific physical realization — an actual Hilbert space, actual expectation values — needs a state. Nothing in §1 or §2 forces one to exist, still less forces which kind.

[COMMITMENT, not forced] This project works with regular, quasi-free, KMS (thermal-equilibrium) states specifically. This is a choice, not a consequence of $(\mathbb{S}, \tau)$: the general space of covariances compatible with the fixed (σ, J) is infinite-dimensional, and nothing rules out non-equilibrium, non-Gaussian states that never reduce to a single scalar. Restricting to the equilibrium family is retained as scaffolding — useful, tractable, and unproven as a necessity.

[DEF] The **family** is the set $\{\omega_\beta : \beta \in (0, \infty)\}$ of states satisfying the commitment above — regular, quasi-free, KMS at some finite β . A state is **admissible** if it is a genuine member of this family. Neither boundary is admissible: the vacuum is an unattained limit of the family (below), and the tracial state is not the same kind of object as its members (below).

[FACT, proven] Given that commitment, the covariance is $\nu(\beta) = \frac{1}{2} \coth(\beta k/2)$ per mode (k the mode frequency, already fixed by (σ, J, g)). β does not modify (σ, J, g) ; it reweights the one fixed geometry, uniformly across every mode, and this reweighting is symplectically invariant — no squeeze transformation undoes it. The vacuum ($\beta \rightarrow \infty, \nu \rightarrow \frac{1}{2}$) is the one point adding nothing beyond the geometric minimum; it is a limit of the family, not a member of it. The opposite limit ($\beta \rightarrow 0$, the tracial state) is not even the same *kind* of object — no covariance operator, no continuous field structure in its GNS representation — so the family’s two edges differ categorically, not just in degree.

[FACT, proven] Distinct β , given a nontrivial dynamics, generate mutually disjoint representations: a standard structural theorem (Bratteli–Robinson, *Operator Algebras and Quantum Statistical Mechanics II*, §5.3), not a covariance estimate. The mechanism: each ω_β ’s modular flow is $\sigma_t^{\omega_\beta} = \alpha_{-\beta t}$ for one fixed dynamics α ; if two such states were quasi-equivalent, their modular flows could differ only by an inner perturbation, forcing α itself to be inner up to rescaling — false for a genuinely outer, ergodic dynamics. The one exception is the degenerate case where α is trivial (the tracial state, already excluded above). So: within the equilibrium commitment, one real number, β , remains — and different

values of it are not different descriptions of one thing, they are different, walled-off worlds.

[COMMITMENT, motivated, not proven] The $\beta \rightarrow \infty$ (vacuum-like) end is placed, specifically, toward the direction later identified as the past — not the future, not both alike. This is a genuine assumption, not a proof: a zero-parameter boundary condition, defended by analogy to a respected proposal in ordinary cosmology (the idea, associated with Roger Penrose, that the very early universe’s gravitational structure was exceptionally smooth, argued from symmetry rather than derived). It names a choice between two provably distinct ends, not a further consequence of anything above.

[SCOPE] β ’s own value is not fixed by $(\mathbb{S}, \tau)$ or by the equilibrium commitment. It is external input, exactly as much as the equilibrium commitment itself is. Neither is claimed to be derivable at present; both are named as what they are.

4. Open

The mechanism, if any, by which $(\mathbb{S}, \tau)$ could select a specific β rather than leaving it external — checked once already for the most natural candidate (a maximally symmetric functional, which reduces to the tracial state, already excluded above as flowless) and not yet checked for anything else.

Whether the equilibrium commitment itself — working with regular, quasi-free, KMS states rather than the larger space nothing rules out — is eventually justified by something more basic, or remains a standing, acknowledged choice.

The exact relationship between the two hard facts of §2: the Clifford structure lives on $\mathbb{S}$ ’s odd τ -half; the commutator structure lives on the even half. Whether this is more than a coincidence of where each already sits — whether τ ’s own grading is doing more structural work than currently claimed — is not yet examined.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author’s responsibility.